O Mirror, Where Art Thou?

Position retrieval using nested sampling and global minimization

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ABSTRACT

Wavefront sensing reveals that a telescope is misaligned, but not which mirror moved or by how much; inferring the mirror positions themselves turns alignment into a direct correction. We describe a simulation framework for inferring telescope mirror misalignment from multi-field Point Spread Function (PSF) measurements projected onto Circular-Bessel and Zernike basis vectors. Two case studies are presented: a Hubble Space Telescope model with five secondary-mirror degrees of freedom and a three-mirror anastigmat with ten primary and secondary mirror degrees of freedom. In both cases, field-dependent multivariate Gaussian likelihoods are explored with nested sampling and refined with differential evolution. Recovered states are validated with residual PSF images and optical path difference (OPD) measurements. In the Hubble benchmark, differential evolution recovers injected misalignments to $\lesssim 0.01\ \mu$m and reduces on-axis OPD error from ~ 2.3 waves to $\sim 3\times10^{-8}$ waves. In the TMA study, the posterior shows broad marginals and correlated parameter pairs that suggest partial degeneracies at higher dimension; differential evolution recovers the truth alignment state less precisely than in the Hubble benchmark ($\lesssim 17\ \mu$m) and reduces on-axis OPD error from ~ 8.24 waves to $\sim 2\times10^{-3}$ waves. This generalizable algorithm enables direct sensing of the state of an optical system and automated alignment.

Key words: Raytracing – Bayesian Analysis – Global Minimization

1 INTRODUCTION

Optical observatories depend on precise alignment of telescope optics to maintain image quality. Wavefront aberrations from mirror translations and tilts propagate into field-dependent point spread functions (PSFs). These field-dependent PSFs therefore provide observables sensitive to rigid-body misalignment even when direct wavefront sensing is unavailable or incomplete.

In this work, we implement a simulation-based solution to the inverse problem of estimating the wavefront from observables: given multi-field PSF observations projected onto basis vectors, we estimate the underlying telescope mirror alignment parameters $\boldsymbol{\theta}$. Synthetic PSFs generated from an open-source ray-tracing model are compared to stored "observations" through a multivariate Gaussian likelihood in coefficient space. We present benchmarks against example two and three mirror telescope models as summarized in Table 1.

Historical HST servicing relied on phase retrieval with focus diversity (Fienup 1982; Löfgren et al. 2014; Krist & Burrows 1997) rather than direct PSF-to-alignment inversion leveraging a full ray-trace model. The commissioning of the James Webb Space Telescope (JWST), a deployable three-mirror anastigmat (Knight et al. 2012), is the most relevant recent precedent. Because the segmented primary can compensate the on-axis wavefront error caused by a misplaced secondary mirror, secondary-mirror misalignment appears mainly in field-dependent terms (astigmatism and focal-plane tilt) that can only be sensed at multiple field points (Acton & Knight 2012). The final stage of JWST telescope commissioning, Multi-Instrument Multi-Field (MIMF) alignment, measured the wavefront with phase retrieval at field points across the science instruments to verify and refine the secondary-mirror position (Acton et al. 2022; McElwain et al. 2023). That approach reconstructs the wavefront at each field point and maps it to mirror corrections through a linear optical model; here we instead infer rigid-body mirror positions directly from PSF intensity with a non-linear ray-trace forward model, obtaining a posterior over the alignment parameters.

Modern open-source ray tracers and nested samplers make it practical to evaluate a full forward model at every likelihood call over a large enough parameter space for the method to be practically useful. Missions such as the Habitable Worlds Observatory (HWO; NASA Goddard Space Flight Center 2023) and the Lazuli Space Observatory (Roy et al. 2026) will require precise and robust alignment algorithms for complex three-mirror anastigmat designs. We investigate the feasibility of solving for telescope misalignments directly from PSF observables as a method for future observatories, like HWO, to rapidly and precisely stabilize their mirrors.

2 METHODS

Misalignments produce aberrations, but phase retrieval recovers only the wavefront; inferring the positions of the telescope optics requires a second, model-based step, such as the linear multi-field inversion used during JWST commissioning (Section 1). In this work, we

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Table 1. Summary of the two telescope models used in this study.

Telescope	Degrees of freedom	Noise	Basis (modes)	Fields	Wavelength
Hubble	5	No	Circular-Bessel (15)	5	0.55 μm
TMA	10	Yes	Zernike (11)	10	0.65 μm

instead use a ray-trace model of the telescope to infer mirror misalignments directly from PSF observations.

To perform misalignment inference, we construct a pipeline (Figure 1) that does the following:

(i) Build a parametric telescope in Optiland (Harrison 2026) and propagate monochromatic light to generate a synthetic PSF.
(ii) Project PSF onto a basis.
(iii) Model observations as multivariate Gaussians in coefficient space with field-specific covariances $\mathbf{\Sigma}_f$ from Monte Carlo calibration.
(iv) Explore the posterior with Nautilus (Lange 2023), and then refine a point estimate with differential evolution.
(v) Validate by comparing nominal and residual-alignment systems in PSF and OPD space.

2.1 Nested sampling

Nested sampling (Skilling 2006) is a Bayesian Monte Carlo method that estimates the evidence $\mathcal{Z} = \int \mathcal{L}(\boldsymbol{\theta})\pi(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta}$ (where $\mathcal{L}(\boldsymbol{\theta})$ is the likelihood and $\pi(\boldsymbol{\theta})$ is the prior) while producing weighted posterior samples. A set of K live points is maintained in prior space and the point with the lowest likelihood is iteratively replaced by a new draw from the prior subject to a rising likelihood threshold. Discarded points contribute to $\ln \mathcal{Z}$ and surviving points approximate $\pi(\boldsymbol{\theta} \mid \text{data})$.

We use `Nautilus`, which implements importance nested sampling (Feroz et al. 2019) with neural-network proposals for efficient exploration of the prior. The prior is encoded by a transform `ptform` that maps uniform unit-cube coordinates $\mathbf{u} \in [0,1]^d$ to physical box bounds on each component of $\boldsymbol{\theta}$. The function passed to the sampler returns $\ln \mathcal{L}(\boldsymbol{\theta})$ only; the prior density is uniform in the box and zero elsewhere. A high-level description of nested sampling is presented in Algorithm 1.

Algorithm 1 Nested sampling algorithm

```
Draw K live points from the prior
while stopping criterion not met do
    Compute the minimum likelihood L_min from the current set of live points
    Add the k^th live point associated with L_min to a list of dead points
    Sample a new point from the prior with likelihood > L_min
    Replace the k^th live point with the newly sampled point
    Evaluate stopping criterion
end while
Sequentially discard remaining live points to complete the evidence estimate
```

The log-likelihood sums independent multivariate Gaussian likelihoods over the normalized field index $f \in \mathcal{F}$. For the TMA, $\mathcal{F} = \{(0, 0.5965), (1,0), (0,1), (1,1), (0,-1), (-1,0), (1,-1), (-1,1), (-1,-1), (0.5, 0.5)\}$, where the near-axis field $(0, 0.5965)$ is used in place of $(0,0)$; the Hubble field set is given in Section 3.1. The log-likelihood is given by equation (1):

$$\ln \mathcal{L}(\boldsymbol{\theta}) = \sum_{f \in \mathcal{F}} \Big\{ -\frac{1}{2} \left[\mathbf{c}_f(\boldsymbol{\theta}) - \mathbf{c}_{f,\mathrm{obs}}\right]^{\top} \mathbf{\Sigma}_f^{-1} \left[\mathbf{c}_f(\boldsymbol{\theta}) - \mathbf{c}_{f,\mathrm{obs}}\right] - \frac{N_z}{2} \ln(2\pi) - \frac{1}{2} \ln\det \mathbf{\Sigma}_f \Big\} \tag{1}$$

where $\mathbf{c}_f(\boldsymbol{\theta})$ is the forward-model basis-coefficient vector at field f, $\mathbf{c}_{f,\mathrm{obs}}$ is the synthetic observation, and N_z is the number of retained modes (15 for Hubble, 11 for the TMA), and $\mathbf{\Sigma}_f$ is the field-specific covariance matrix.

2.2 Differential evolution

Differential evolution (DE) is a population-based global optimizer that refines a `Nautilus`-derived starting point into a single best fit misalignment vector by minimizing coefficient-space residuals. It was introduced by Storn & Price (1997) and is implemented in `scipy.optimize` as `differential_evolution`. DE maintains a population of candidate parameter vectors $\{\theta_1, \ldots, \theta_N\}$ inside box bounds. For each generation, it iteratively improves the candidate solution as follows:

(i) Pick a target individual θ_i.
(ii) Build a trial vector θ_{trial} by combining traits from other population members.
(iii) Evaluate the objective at θ_{trial}.
(iv) Keep θ_{trial} if it is better than θ_{i}; otherwise keep θ_{i}.

Over many generations, the population migrates toward low objective values. DE is stochastic, can escape shallow local minima better than single-start local methods, and is a natural second stage after nested sampling has already narrowed the search region. After `Nautilus`, we start DE from the weighted median of the final 100 posterior samples and search within bounds set by the 2.1%–97.9% weighted quantiles of the same samples rather than the full prior box. This starting point is distinct from the full-posterior marginal medians shown in the corner plots (Figures 4 and 6). In this work, we further polish the DE result by performing nonlinear optimization using the best DE point as a starting guess for the L-BFGS-B optimizer available through the `scipy.optimize` module's `minimize` method.

3 APPLYING TO TELESCOPE MODELS

3.1 Hubble secondary alignment

As an example of a 2-mirror system, we model a Ritchey–Chrétien-style Hubble Space Telescope in Optiland (Figure 2; EPD = 2400 mm, $\lambda = 0.55\,\mu$m) with misalignments applied only to the secondary mirror. The five degrees of freedom are: decenter sdx, sdy and despace sdz (microns), and tilts stx, sty (degree). Normalized field coordinates are relative to a maximum field angle of 0.038°.

PSFs are computed with `MMDFTPSF` at `grid_size=64`, cropped to 64×64 pixels, and projected onto 15 circular-Bessel basis functions. Observations are noise free. Synthetic truth coefficients are stored at five normalized fields: $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$, and $(-1,-1)$.

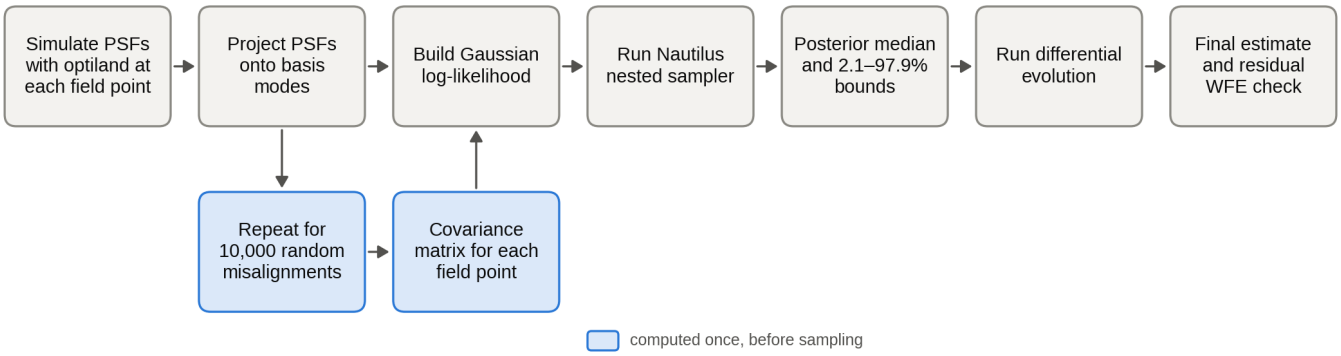


Figure 1. Inference pipeline: Optiland forward model, basis-coefficient feature extraction, multivariate Gaussian likelihood, Nautilus nested sampling, and differential-evolution refinement with PSF/OPD validation.

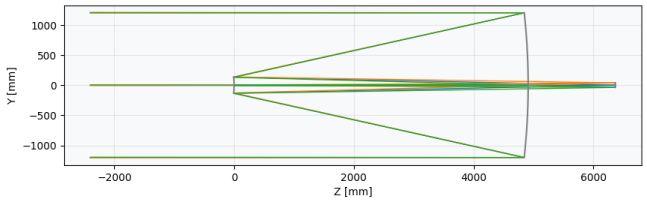


Figure 2. Optiland ray trace of the Hubble model. Likelihood evaluation uses circular-Bessel coefficients at five field coordinates (see text).

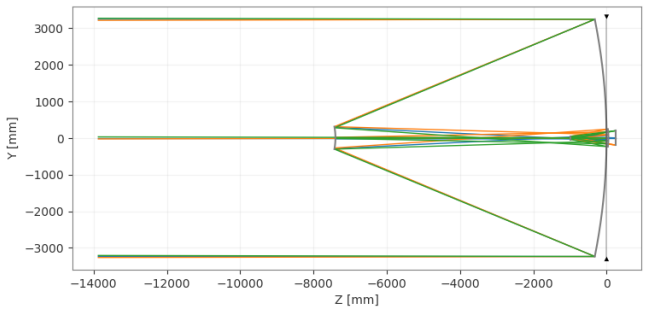


Figure 3. Raytrace of the TMA model in Optiland. Multi-field PSF features used in inference are evaluated at ten field coordinates across the focal plane (see text).

For each field, a 15×15 covariance matrix is estimated from 10^4 Monte Carlo draws of $\boldsymbol{\theta}$ under zero-mean Gaussian perturbations ($\sigma = 100\,\mu$m on translations, $\sigma = 0.167$ degrees on tilts), with each draw rounded to two decimal places.

Uniform priors on the unit cube map to $\pm 200\,\mu$m on (sdx, sdy, sdz) and ± 0.167 degrees on (stx, sty); samples from the prior transform are rounded to $0.01\,\mu$m and 0.01 degrees. Nautilus runs with `n_dim=5`. In our completed benchmark, the sampler performed 42 800 likelihood evaluations with $\ln \mathcal{Z} \approx +107.5$ and $N_{\mathrm{eff}} \approx 1.43 \times 10^4$. The weighted median of the last 100 posterior samples provides the starting point for DE.

The DE objective minimizes $\sum_{k=1}^{15} [c_k(\boldsymbol{\theta}) - c_{k,\mathrm{obs}}]^2$ at the on-axis field only. Multi-field terms are not included in the DE stage. Nautilus constrains sdz and the tilts but leaves sdx and sdy broad (Section 4.1); in the noise-free case, the on-axis coefficients alone are sufficient for DE to recover all five parameters within the posterior-derived bounds.

3.2 Three mirror anastigmat

We apply the same inference pipeline to a compact three-mirror anastigmat (TMA) based on the optical design described in Kim et al. (2023), implemented in Optiland as a four-mirror prescription (Figure 3) with entrance-pupil diameter EPD = 6471.26 mm and monochromatic wavelength $\lambda = 0.65\,\mu$m. Unlike the Hubble case, misalignments are applied to both the primary and secondary mirrors: ten rigid-body degrees of freedom parameterize decenter (x, y), despace (z), and tilt (x, y) on each mirror. Translations are specified in microns and tilts are in degrees. Normalized field coordinates are relative to a maximum field angle of 0.285°.

For each trial alignment vector $\boldsymbol{\theta}$, we build the TMA model and propagate the field with `MMDFTPSF` using 512 rays, and retain a central 256×256 pixel intensity patch. The patch is projected onto the first 11 Zernike modes, yielding an 11-element coefficient vector $\mathbf{c}_f(\boldsymbol{\theta})$ at each field point f. The synthetic observations $\mathbf{c}_{f,\mathrm{obs}}$ are computed from PSFs with additive Gaussian noise ($\sigma = 0.01$ per pixel, in PSF intensity units), while forward-model PSFs in the likelihood are noise free. Each covariance $\boldsymbol{\Sigma}_f$ is estimated from 10^4 noise-free Monte Carlo draws with $\sigma = 250\,\mu$m on (pdx, pdy), $50\,\mu$m on pdz, $800\,\mu$m on (sdx, sdy), $100\,\mu$m on sdz, 0.003 degrees on (ptx, pty), and 0.13 degrees on (stx, sty), with each draw rounded to two decimal places.

The prior transform maps uniform unit-cube samples to independent uniform boxes: primary decenters (pdx, pdy) and despace $pdz \in [-800, 800]\,\mu$m, secondary decenters (sdx, sdy) and despace $sdz \in [-2500, 2500]\,\mu$m, primary tilts $\in [-0.01, 0.01]$ degrees, and secondary tilts $\in [-0.4, 0.4]$ degrees. The weighted median $\mathbf{x}_0$ and the 2.1%–97.9% weighted quantiles of the final 100 posterior samples initialize and bound differential evolution. The DE objective is the unweighted sum of squared coefficient residuals over all ten fields, $\sum_{f \in \mathcal{F}} \sum_{k=1}^{11} [c_{f,k}(\boldsymbol{\theta}) - c_{f,k,\mathrm{obs}}]^2$, minimized with `tol=1e-10`; the run reached the limit of 2000 generations (final objective 2.2×10^{-7}) before L-BFGS-B polishing.

Recovered alignments $\mathbf{c}$ are validated by constructing a residual telescope with misalignments $(\boldsymbol{\theta}_{\mathrm{true}} - \mathbf{c})$ and comparing on-axis PSF images and OPD maps against the nominal system via `OPD` at field $(0, 0)$. We report both per-component residuals $\boldsymbol{\theta}_{\mathrm{true}} - \mathbf{c}$ and the RMS of the OPD difference in waves.

Both studies use Optiland for ray tracing, Nautilus for nested sam-

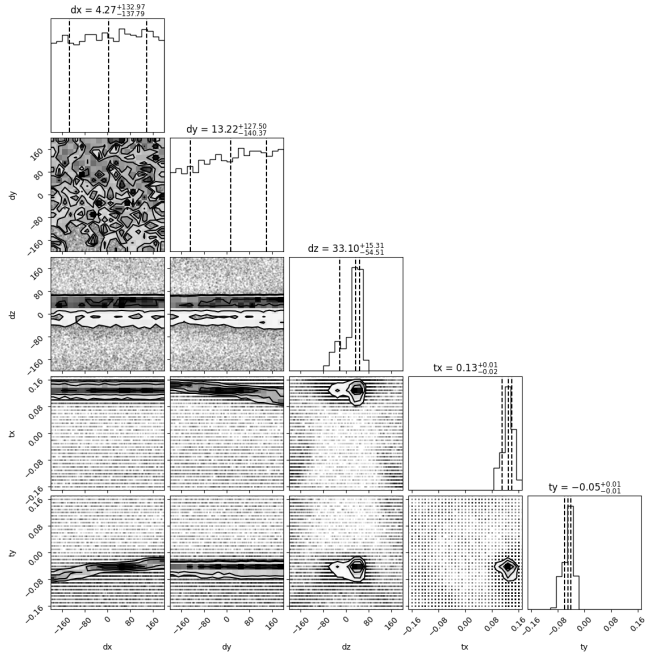


Figure 4. Nautilus posterior corner plot for the five secondary-misalignment parameters of the Hubble model (axes labelled dx, dy, dz, tx, ty correspond to sdx, sdy, sdz, stx, sty). Titles give the full-posterior marginal medians.

pling, and SciPy (Virtanen et al. 2020) differential evolution for final refinement; the TMA study uses POPPY (Perrin et al. 2012) for the Zernike basis.

4 RESULTS

We evaluated both pipelines in simulation using analysis notebooks that record truth misalignments $\boldsymbol{\theta}_{\rm true}$, nested-sampling posteriors, DE solutions $\mathbf{c}$, and validation metrics. The Hubble benchmark projects PSFs onto circular-Bessel basis functions, while the TMA study uses Zernike polynomials from POPPY. The two bases weight the PSF very differently. Zernike polynomials are orthogonal on the unit disk and grow toward its edge, so projecting a PSF intensity onto the first few modes gives, in effect, low-order image moments: piston measures the total flux, tip and tilt the centroid, defocus the radial width, astigmatism the ellipticity, and coma the third-order asymmetry. Because these weights rise with radius and are nearly flat across a compact core, Zernike features respond well to broad, strongly aberrated PSFs but give heavy weight to the faint, noise-dominated wings and are largely blind to the ring structure of a nearly diffraction-limited PSF. The circular-Bessel functions, $J_{n+1}(2\pi\rho)\cos(m\phi)/\rho$ and $J_{n+1}(2\pi\rho)\sin(m\phi)/\rho$, are the Fourier transforms of the Zernike polynomials (Noll 1976), so their radial structure follows the diffraction rings of a circular pupil (the lowest-order term is the Airy amplitude) and their weight decays away from the core. Circular-Bessel features should therefore be more sensitive to small misalignments and less affected by noise in the wings, at the cost of a radial scale that must be matched to λ/D in detector pixels and a smaller capture range for large misalignments that scatter light far from the core. A Zernike basis may therefore suit a coarse search from a strongly misaligned state, and a circular-Bessel basis the final refinement; a controlled comparison of the two on the same model and noise level is left to future work.

Table 2. Hubble secondary: injected truth and DE-recovered misalignment (noise-free benchmark). Tilts are in degrees.

Parameter	True value	Recovered (DE)	Units
sdx	52.22	52.22	μm
sdy	−113.86	−113.86	μm
sdz	38.99	38.99	μm
stx	0.13	0.13	degree
sty	−0.04	−0.04	degree

4.1 Hubble benchmark

One noise-free realization was injected with translations $(sdx, sdy, sdz) = (52.22, -113.86, 38.99)\,\mu$m and tilts $(stx, sty) = (0.13, -0.04)$ degrees. Nautilus ($\ln\mathcal{Z} \approx +107.5$) tightly constrained both tilts and, more loosely, sdz, but left sdx and sdy unconstrained: their marginals in Figure 4 are close to flat across the $\pm 200\,\mu$m prior. The weighted median of the final 100 posterior samples, used to start DE, was (131.13, 56.64, 40.87, 0.12, −0.03), and the corresponding 2.1%–97.9% DE bounds on sdx and sdy each span $\approx$ 310–320 μm. Differential evolution converged to $\mathbf{c}$ = (52.22, −113.86, 38.99, 0.130, −0.040) with final objective $\lesssim 10^{-15}$, yielding sub-0.01 μm residuals on all translations (Table 2).

With no correction, on-axis OPD RMS is $\sim$ 2.26 waves. Using the DE starting point (the median of the final 100 posterior samples) alone reduces this to $\sim$ 0.12 waves. After DE refinement, on-axis OPD RMS falls to $\sim 3 \times 10^{-8}$ waves. Figure 4 shows posterior marginals for all five secondary misalignments, and Figure 5 shows the on-axis PSF and RMS wavefront error at each stage of the refinement.

4.2 TMA benchmark

The TMA study uses the noisy forward model described in Sec. 3.2. Ten-dimensional nested sampling is computationally expensive; posterior medians need not coincide with injected truth when the likelihood contains flat directions. The posterior marginals in Figure 6 suggest partial degeneracies in the ten-dimensional problem. The primary decenter pdx and both primary tilts are close to flat across their prior ranges, the secondary decenters remain broad, and several pairs are strongly correlated, notably pdz–sdz, ptx–sdy and pty–sdx. The tilt–decenter correlations have a familiar optical origin: a tilt of one mirror about the x (y) axis and a decenter of the other along y (x) both introduce coma that is nearly constant across the field, so one can largely cancel the coma introduced by the other. The same mechanism made secondary-mirror misalignment on JWST detectable only through field-dependent aberrations (Section 1): a compensating motion of another mirror cancels the field-constant error and leaves a field-dependent residual, which is why our likelihood combines several field points. The pdz–sdz correlation similarly reflects the trade-off between the two despaces in setting focus. Only the secondary tilts are tightly constrained. Distinct alignment states may therefore produce similar Zernike vectors, and the likelihood alone may not determine every parameter uniquely.

Nautilus performed 93 200 likelihood evaluations with $\ln\mathcal{Z} \approx +484.7$ and $N_{\rm eff} \approx 2.04 \times 10^4$. With no correction, on-axis OPD RMS is $\sim$ 8.24 waves, and the DE starting point $\mathbf{x}_0$ (the median of the final 100 posterior samples) alone reduces this to $\sim$ 0.99 waves. Table 3 reports the truth and DE-refined alignment states: the refinement reduces on-axis OPD RMS to $\sim 2 \times 10^{-3}$ waves while leaving residuals of up to $\sim$ 16.5 μm on the translations (largest on secondary decenter sdy). Primary tilts are near zero at truth and

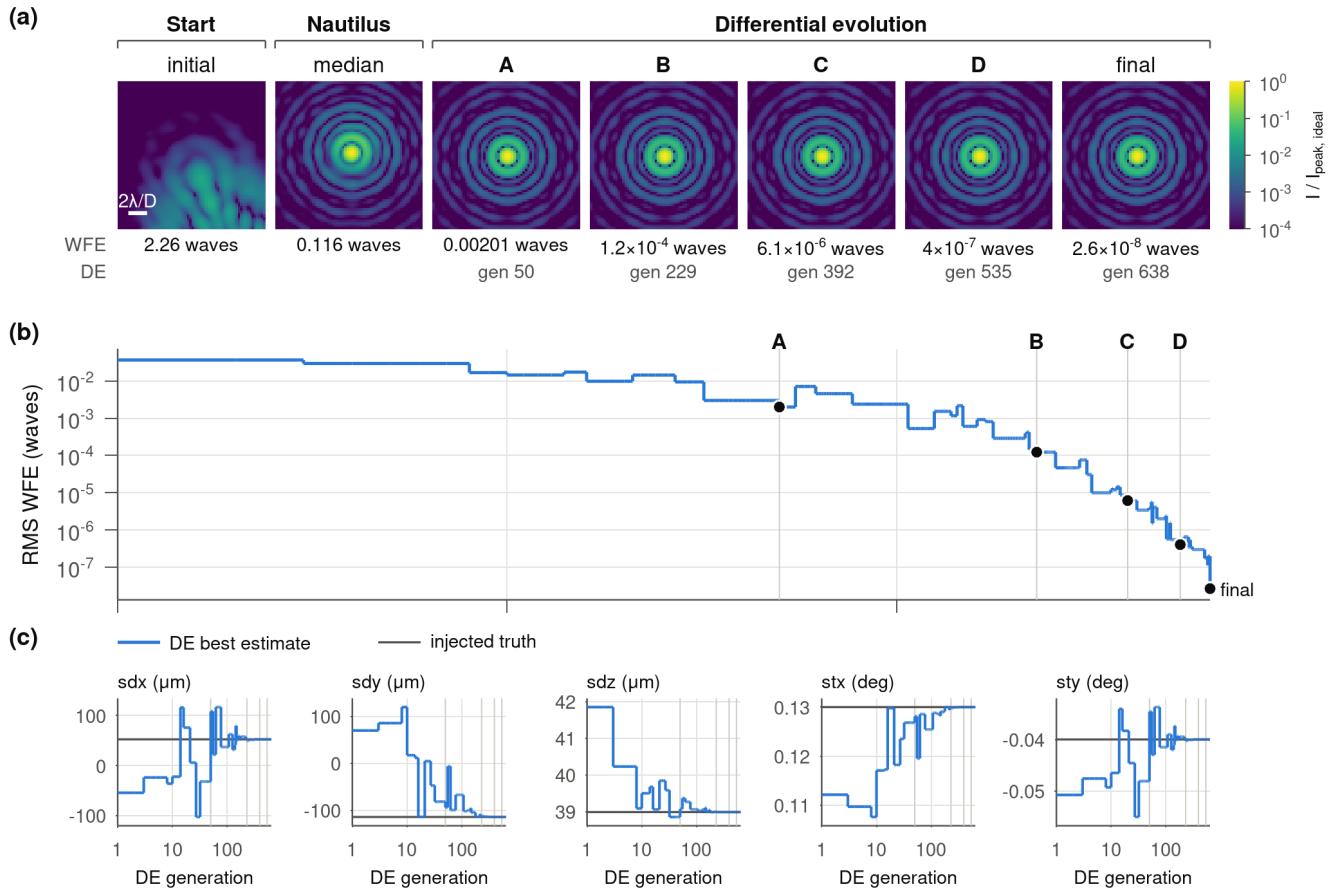


Figure 5. Hubble benchmark: on-axis PSF and RMS wavefront error (WFE; the on-axis OPD RMS quoted in the text) as the alignment is refined. (a) PSFs for the uncorrected system, the Nautilus starting point (median of the final 100 posterior samples), intermediate DE solutions and the final DE solution. (b) RMS WFE of the DE best estimate versus generation. (c) DE best estimate of each secondary-mirror parameter versus generation, with the injected truth.

Table 3. TMA: injected truth and DE-recovered alignment.

Parameter	True value	Recovered	Units
pdx	−152.75	−159.43	μm
pdy	316.91	309.97	μm
pdz	26.63	24.77	μm
sdx	−42.31	−38.01	μm
sdy	923.33	939.85	μm
sdz	−32.75	−34.56	μm
ptx	2.98×10^{-5}	2.22×10^{-4}	degree
pty	1.61×10^{-5}	6.80×10^{-5}	degree
stx	0.0269	0.0272	degree
sty	0.0098	0.0098	degree

remain poorly determined relative to translations. Figure 7 shows the on-axis PSF and RMS wavefront error at each stage of the refinement.

5 CONCLUSION

We presented a Bayesian pipeline for inferring telescope rigid body misalignment from simulated multi-field observations for optics with known prescriptions. Nested sampling with Nautilus provides global posterior exploration and credible regions; differential evolution delivers a high-precision point estimate seeded by the median of the final posterior samples. In a noise-free Hubble secondary-mirror benchmark, the method recovers five degrees of freedom to $\lesssim 0.01\,\mu$m and reduces on-axis OPD error from ~ 2.3 waves to $\sim 3 \times 10^{-8}$ waves. In a ten-degree-of-freedom TMA study with noisy PSFs, DE recovered the translations to $\lesssim 17\,\mu$m and reduced on-axis OPD error from ~ 8.24 waves to $\sim 2 \times 10^{-3}$ waves. This approach, while computationally intensive allows measurement of the state of an optical system with a single measurement of a wide-field imaging instrument as is typical of modern TMA observatories and is expected to enable deterministic alignment of optical systems.

Future work includes extending the pipeline to chromatic and realistic noise models and applying the pipeline to laboratory or on-sky PSF campaigns relevant to Habitable Worlds Observatory class TMA designs.

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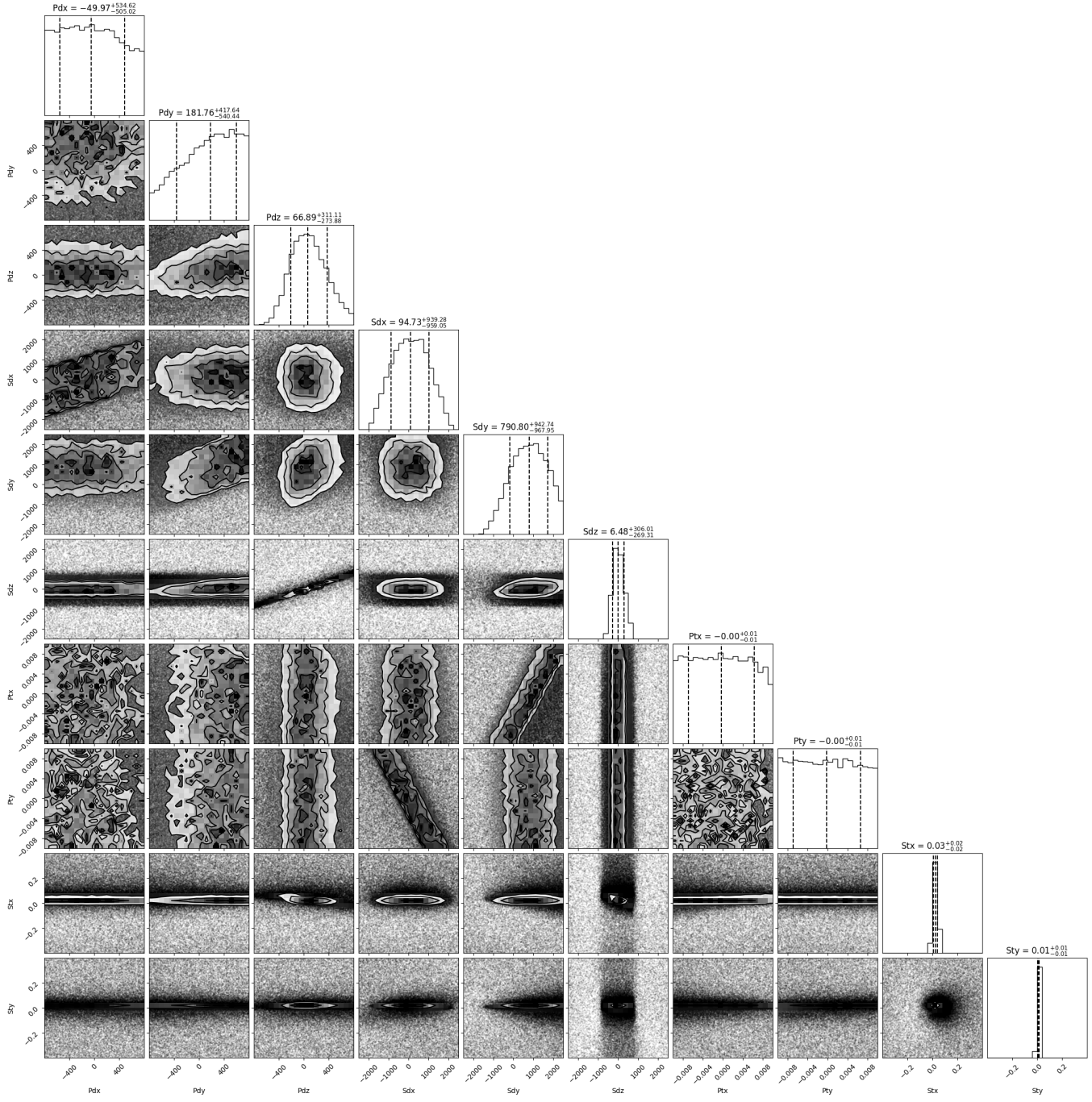


Figure 6. Nautilus posterior corner plot for the ten primary/secondary misalignment parameters of the TMA model.

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DATA AVAILABILITY

The Jupyter notebooks containing code, analysis, and figures used in this work are available upon request (or put in a github repo?)

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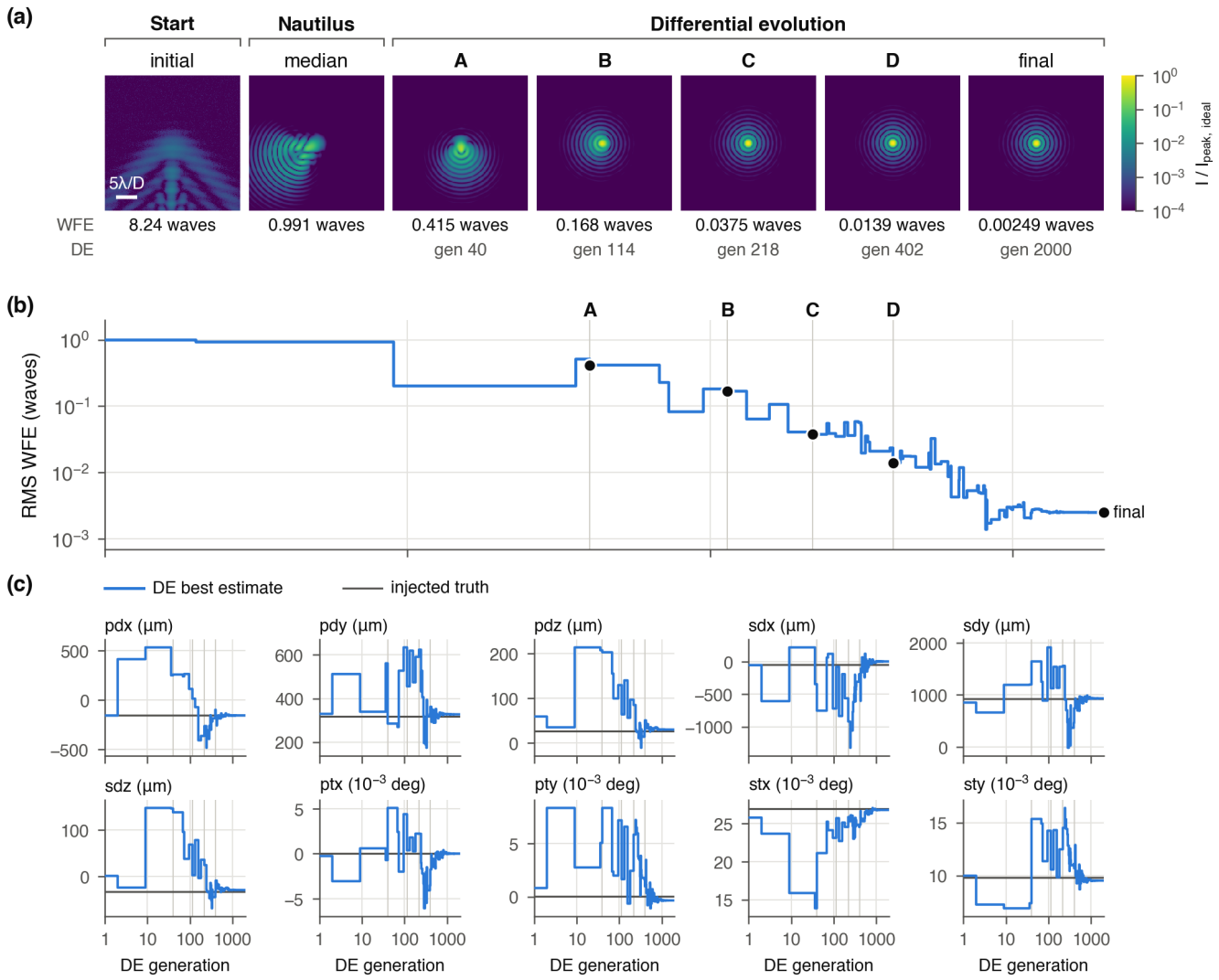


Figure 7. TMA benchmark: (a) On-axis PSFs at $\lambda = 0.65\,\mu$m: the observed misaligned PSF (Gaussian noise, $\sigma = 0.01\%$ of the ideal peak), the Nautilus starting point (median of the final 100 posterior samples), during differential evolution (DE) at generations 40, 114, 218 and 402 (A–D), and after 2000 generations (final). Intensity is relative to the aligned system's peak, so each panel's peak equals its Strehl ratio (log stretch, floor 10^{-4}). Values are RMS wavefront error in waves, the RMS of the OPD difference from the nominal design. (b) RMS WFE of the best DE member at each generation; letters mark the PSFs in (a). (c) Best-member value of each alignment parameter at each generation (blue) and the injected truth (gray).

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